Lecture 11

Universal enveloping algebras and PBW

Objectives. *A representation turns brackets into commutators of operators. We build the associative algebra that does this universally. Then we prove that its ordered monomials form a basis.*

By the end of this lecture you will be able to:

- Construct $U(\mathfrak{g})$ and pass between $\mathfrak{g}$-modules and $U(\mathfrak{g})$-modules.
- Prove the Poincaré–Birkhoff–Witt theorem, including the step that uses the Jacobi identity.
- Identify $\operatorname{gr} U(\mathfrak{g})$ with $\operatorname{Sym}(\mathfrak{g})$ and factor $U(\mathfrak{g})$ along a triangular decomposition.
- Reorder words in $U(\mathfrak{sl}_2)$ and compare the result in the associated graded algebra.

Lecture 6 built the irreducible $\mathfrak{sl}_2$-modules from explicit formulas. We need a construction that works for every complex semisimple Lie algebra $\mathfrak{g}$. Choose a Cartan subalgebra $\mathfrak{h}$ and positive roots as in Lecture 9, and let $\mathfrak{n}_+$ be the sum of the positive root spaces, the positive-root subalgebra. Fix a linear functional λ on $\mathfrak{h}$. We seek a module generated by one vector v with $hv = \lambda(h)v$ for $h \in \mathfrak{h}$ and $\mathfrak{n}_+ v = 0$, imposing no further relations. The algebra $U(\mathfrak{g})$ of this lecture lets us express those relations. The PBW theorem will show that the resulting module is nonzero. This paragraph is a preview of Lecture 12, and nothing in this lecture depends on it.

Throughout, $\mathfrak{g}$ is a finite-dimensional complex Lie algebra. An *algebra* is an associative complex algebra with 1, and algebra homomorphisms preserve 1. This lecture departs from the standing convention on dimension: tensor algebras, polynomial algebras, $U(\mathfrak{g})$ and their modules are infinite-dimensional in general. All arguments are algebraic.

11.1 The enveloping algebra

For a vector space V, the *tensor algebra* is

$$T(V) = \bigoplus_{k \geq 0} V^{\otimes k}, \qquad V^{\otimes 0} = \mathbb{C},$$

with product given on pure tensors by concatenation. Multilinear algebra gives its universal property. A linear map $\varphi \colon V \to A$ into an algebra extends uniquely to an algebra homomorphism $T(V) \to A$, given by

$$v_1 \otimes \cdots \otimes v_k \longmapsto \varphi(v_1) \cdots \varphi(v_k).$$

Every algebra A is a Lie algebra under $[a, b] = ab - ba$. The Jacobi identity holds by the expansion in Lecture 1.

Definition 11.1 (Universal enveloping algebra)**.** Let I be the two-sided ideal of $T(\mathfrak{g})$ generated by the elements

$$x \otimes y - y \otimes x - [x, y] \qquad (x, y \in \mathfrak{g}).$$

The *universal enveloping algebra* is $U(\mathfrak{g}) = T(\mathfrak{g})/I$. Let $\iota\colon \mathfrak{g} \to U(\mathfrak{g})$ be the inclusion $\mathfrak{g} = \mathfrak{g}^{\otimes 1} \subseteq T(\mathfrak{g})$ followed by the quotient map.

By construction,

$$\iota(x)\iota(y) - \iota(y)\iota(x) = \iota([x, y]) \quad \text{for all } x, y \in \mathfrak{g}.$$

So ι is a Lie algebra homomorphism into $U(\mathfrak{g})$ with its commutator bracket. Injectivity of ι is not yet known. It will follow from PBW.

Proposition 11.2 (Universal property)**.** *Let A be an algebra and $\varphi\colon \mathfrak{g} \to A$ a linear map with*

$$\varphi([x, y]) = \varphi(x)\varphi(y) - \varphi(y)\varphi(x) \quad \textit{for all } x, y \in \mathfrak{g}.$$

There is a unique algebra homomorphism $\tilde{\varphi}\colon U(\mathfrak{g}) \to A$ with $\tilde{\varphi} \circ \iota = \varphi$.

Proof. Let $\Phi\colon T(\mathfrak{g}) \to A$ extend φ. On a generator of I,

$$\Phi(x \otimes y - y \otimes x - [x, y]) = \varphi(x)\varphi(y) - \varphi(y)\varphi(x) - \varphi([x, y]) = 0.$$

The kernel of Φ is a two-sided ideal, so it contains I. Hence Φ factors through $U(\mathfrak{g})$.

The algebra $U(\mathfrak{g})$ is generated by $\iota(\mathfrak{g})$, because $T(\mathfrak{g})$ is generated by $\mathfrak{g}$. Two homomorphisms that agree on $\iota(\mathfrak{g})$ are therefore equal. ∎

Proposition 11.3 (Modules)**.** *Let V be a vector space of any dimension, and let $\mathfrak{gl}(V)$ denote $\mathrm{End}(V)$ with the commutator bracket. The assignments $\rho \mapsto \tilde{\rho}$ and $\pi \mapsto \pi \circ \iota$ are inverse bijections between representations $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ and unital $U(\mathfrak{g})$-module structures $\pi\colon U(\mathfrak{g}) \to \mathrm{End}(V)$. A linear map between two modules is a $\mathfrak{g}$-intertwiner exactly when it is a $U(\mathfrak{g})$-module map.*

Proof. A representation is a linear map $\mathfrak{g} \to \mathrm{End}(V)$ satisfying the hypothesis of Proposition 11.2. That proposition gives $\tilde{\rho}$ with $\tilde{\rho} \circ \iota = \rho$. Conversely, $\pi \circ \iota$ preserves brackets because ι and π do. Uniqueness in Proposition 11.2 gives $\widetilde{\pi \circ \iota} = \pi$.

Let $T\colon V \to W$ be a linear map between two modules. Put

$$C_T = \{u \in U(\mathfrak{g}) : T\pi_V(u) = \pi_W(u)T\}.$$

The set C_T is a linear subspace, since its defining condition is linear in u. It contains 1 and is closed under products. For $u, u' \in C_T$,

$$T\pi_V(uu') = T\pi_V(u)\pi_V(u') = \pi_W(u)T\pi_V(u') = \pi_W(u)\pi_W(u')T.$$

So C_T is a subalgebra. It contains $\iota(\mathfrak{g})$ exactly when T is a $\mathfrak{g}$-intertwiner. In that case $C_T = U(\mathfrak{g})$, since $\iota(\mathfrak{g})$ generates $U(\mathfrak{g})$. ∎

In calculations we write x for $\iota(x)$. Until Corollary 11.9 this is only notation.

11.2 The filtration and spanning

Let U_p be the image of $\bigoplus_{k\le p}\mathfrak{g}^{\otimes k}$ in $U(\mathfrak{g})$. It is spanned by products of at most p elements of $\mathfrak{g}$. Every generator of I has positive tensor degree. The sum $\bigoplus_{k\ge 1}\mathfrak{g}^{\otimes k}$ is a two-sided ideal, so it contains I, and $1\notin I$. Put $U_{-1}=0$. Then

$$\mathbb{C}=U_0\subseteq U_1\subseteq U_2\subseteq\cdots,\qquad \bigcup_p U_p=U(\mathfrak{g}),\qquad U_pU_q\subseteq U_{p+q}.$$

Fix an ordered basis $(x_1,\dots,x_n)$ of $\mathfrak{g}$. A nondecreasing sequence $\Sigma=(i_1\le\cdots\le i_p)$ has *length* p. Its *ordered monomial* is

$$x_\Sigma=x_{i_1}x_{i_2}\cdots x_{i_p}\in U(\mathfrak{g}),$$

with $x_\varnothing=1$. For an index λ, write $\lambda\le\Sigma$ if $\lambda\le i_1$. This holds for every λ when $\Sigma=\varnothing$.

Lemma 11.4 (Reordering). *Let $y_1,\dots,y_k\in\mathfrak{g}$ and $\sigma\in S_k$. Then*

$$y_{\sigma(1)}\cdots y_{\sigma(k)}-y_1\cdots y_k\in U_{k-1}.$$

Proof. Write σ as a product of adjacent transpositions. Consecutive words in the resulting chain differ by swapping two neighbours a,b. With u and v the products of the other $k-2$ factors,

$$u\,ab\,v-u\,ba\,v=u\,[a,b]\,v\in U_{k-1}.$$

Summing these differences along the chain proves the claim. ∎

Proposition 11.5 (Spanning). *The ordered monomials of length at most p span U_p. Hence the ordered monomials span $U(\mathfrak{g})$.*

Proof. Use induction on p. For $p=0$, the space $U_0=\mathbb{C}$ is spanned by $x_\varnothing=1$. By multilinearity, U_p is spanned by U_{p-1} and the words $x_{j_1}\cdots x_{j_p}$. Let Σ be the indices $j_1,\dots,j_p$ sorted into nondecreasing order. By Lemma 11.4,

$$x_{j_1}\cdots x_{j_p}-x_\Sigma\in U_{p-1}.$$

The induction hypothesis covers U_{p-1}. ∎

Theorem 11.6 (Poincaré–Birkhoff–Witt). *The ordered monomials x_Σ form a basis of $U(\mathfrak{g})$.*

Spanning is proved. For independence we build a module on which the ordered monomials act independently.

One example shows why independence needs an argument. Take $\mathfrak{sl}_2$ with the ordered basis (f,h,e) and the rewriting rules

$$eh=he-2e,\qquad hf=fh-2f,\qquad ef=fe+h,$$

which §11.5 derives from the brackets. The word ehf has two out-of-order neighbouring pairs, so reordering can begin in two ways. Figure 11.1 shows both routes. They end at the same ordered expression. If they ended at different ordered expressions, the difference would be a nontrivial linear relation among ordered monomials. This example is a preview and not part of the proof. The general compatibility is proved in §11.3, where it uses the Jacobi identity.

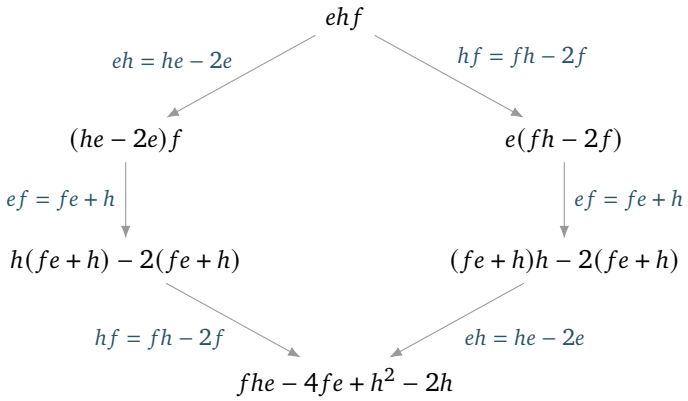


Figure 11.1: Two ways to reorder the word ehf in $U(\mathfrak{sl}_2)$, read from the top down. Each arrow applies the rule printed beside it, and both routes end at the same ordered expression.

A natural place to stop. We have built $U(\mathfrak{g})$, shown that its ordered monomials span it, and stated the PBW theorem. The next part proves that these monomials are independent, by constructing a module on polynomials.

11.3 Independence of ordered monomials

Let $S = \mathbb{C}[z_1, \dots, z_n]$, and let S_p be the polynomials of degree at most p, with $S_{-1} = 0$. For Σ as above put $z_\Sigma = z_{i_1} \cdots z_{i_p}$. The monomials z_Σ of length at most p form a basis of S_p.

We construct operators on the polynomial space S, one for each element of $\mathfrak{g}$. The leading term of the operator of x_λ is multiplication by z_λ. When λ is at most every index of a monomial, the operator only multiplies. When λ is out of order, we swap it with the first index of the monomial, and the swap adds a Lie bracket. These operators form a representation of $\mathfrak{g}$ on S. An ordered monomial x_Σ then sends the polynomial 1 to z_Σ, and the polynomials z_Σ are independent. The commutator relation holds by construction in the ordered cases. The only compatibility left after them is the Jacobi identity.

Lemma 11.7. *There is a bilinear map $\mathfrak{g} \times S \to S$, written $(x, z) \mapsto x \cdot z$, such that for all indices λ, μ and all $p \geq 0$:*

(A) $x_\lambda \cdot z_\Sigma = z_\lambda z_\Sigma$ *whenever* $\lambda \leq \Sigma$*;*

(B) $x_\lambda \cdot z - z_\lambda z \in S_p$ *for every* $z \in S_p$*;*

(C) $x_\lambda \cdot (x_\mu \cdot z) - x_\mu \cdot (x_\lambda \cdot z) = [x_\lambda, x_\mu] \cdot z$ *for every* $z \in S$*.*

By (B) and linearity, $x \cdot z \in S_{p+1}$ for $x \in \mathfrak{g}$ and $z \in S_p$. Condition (C) is linear in x_λ and x_μ. Once it holds on basis vectors, it holds for all pairs of elements of $\mathfrak{g}$.

Before the proof we compute what (A) and (C) force in the smallest case.

Example 11.8 (Degree one in $\mathfrak{sl}_2$)**.** Take the ordered basis (f, h, e), so that $f < h < e$, and write z_f, z_h, z_e for the variables of S. Condition (A) gives $f \cdot 1 = z_f$, $h \cdot 1 = z_h$ and $e \cdot 1 = z_e$. It also gives $f \cdot z_e = z_f z_e$, since $f < e$. Now apply (C) to the pair e, f and the polynomial $z = 1$:

$$e \cdot z_f = e \cdot (f \cdot 1) = f \cdot (e \cdot 1) + [e, f] \cdot 1 = z_f z_e + z_h.$$

In the same way, $[h,f]=-2f$ and $[e,h]=-2e$ give

$$h\cdot z_f = f\cdot z_h + [h,f]\cdot 1 = z_h z_f - 2z_f, \qquad e\cdot z_h = h\cdot z_e + [e,h]\cdot 1 = z_e z_h - 2z_e.$$

The first computation also reads $e\cdot z_f - f\cdot z_e = z_h = [e,f]\cdot 1$, which is (C) on the polynomial 1. Each result is a product of variables plus a lower correction. So the operators of h and e are not multiplication operators. None of the three operators is a derivation of the polynomial algebra, since each sends 1 to a nonzero polynomial.

Proof. We define the action on $\mathfrak{g}\times S_p$ by induction on p, each stage extending the previous one. Stage p must satisfy (A) and (B) for $z\in S_p$, and (C) for $z\in S_{p-1}$. Call these conditions (A_p), (B_p) and (C_p). By the remark above, (C_p) uses only the action on S_p.

Stage 0. Put $x_\lambda\cdot 1 = z_\lambda$. Then ($A_0$) and ($B_0$) hold, and ($C_0$) is empty.

Stage $p\ge 1$: definition. Let z_Σ have length p.

(i) If $\lambda\le\Sigma$, put $x_\lambda\cdot z_\Sigma = z_\lambda z_\Sigma$.

(ii) Otherwise $\Sigma=(\mu,T)$ with $\mu<\lambda$, $\mu\le T$ and T of length $p-1$. By (B_{p-1}),

$$x_\lambda\cdot z_T = z_\lambda z_T + y, \qquad y\in S_{p-1}.$$

Put

$$x_\lambda\cdot z_\Sigma = z_\lambda z_\Sigma + x_\mu\cdot y + [x_\lambda,x_\mu]\cdot z_T.$$

Rule (ii) is the value that (C) demands when x_λ moves past x_μ, as in Example 11.8. The right side of (ii) uses only the action on S_{p-1}.

(A_p) and (B_p). Rule (i) gives (A_p) in length p. In rule (ii), the terms $x_\mu\cdot y$ and $[x_\lambda,x_\mu]\cdot z_T$ lie in S_p by (B_{p-1}). On S_{p-1}, both conditions were proved at stage $p-1$.

(C_p), the ordered case. Keep the notation of rule (ii). The monomial $z_\lambda z_T$ has length p, and $\mu<\lambda$, $\mu\le T$. So μ is at most every index of $z_\lambda z_T$, and rule (i) applies to $x_\mu\cdot(z_\lambda z_T)$:

$$x_\mu\cdot(x_\lambda\cdot z_T) = x_\mu\cdot(z_\lambda z_T) + x_\mu\cdot y = z_\mu z_\lambda z_T + x_\mu\cdot y = z_\lambda z_\Sigma + x_\mu\cdot y.$$

Also $x_\mu\cdot z_T = z_\Sigma$ by (A_{p-1}). Rule (ii) therefore gives

$$x_\lambda\cdot(x_\mu\cdot z_T) = x_\lambda\cdot z_\Sigma = z_\lambda z_\Sigma + x_\mu\cdot y + [x_\lambda,x_\mu]\cdot z_T,$$

that is,

$$x_\lambda\cdot(x_\mu\cdot z_T) = x_\mu\cdot(x_\lambda\cdot z_T) + [x_\lambda,x_\mu]\cdot z_T. \tag{11.1}$$

This is (C_p) for $z=z_T$ of length $p-1$ when $\mu<\lambda$ and $\mu\le T$. Exchanging λ and μ changes the sign of both sides of (C). So (C_p) also holds when $\lambda<\mu$ and $\lambda\le T$. It is trivial when $\lambda=\mu$. For $z\in S_{p-2}$ it is (C_{p-1}).

(C_p), the Jacobi case. Let z_T be an ordered monomial of length $p-1$. For $p=1$, $T=\varnothing$ and an ordered case applies. For $p\ge 2$, let ν be the first index of T. If $\min(\lambda,\mu)\le\nu$, an ordered case applies. It remains to take $p\ge 2$ with $\nu<\lambda$ and $\nu<\mu$. Write $T=(\nu,\Psi)$. Since $\nu\le\Psi$, (A_{p-1}) gives

$$z_T = x_\nu\cdot z_\Psi, \qquad z_\Psi\in S_{p-2}.$$

On z_Ψ, (C_{p-1}) holds for all pairs of elements of $\mathfrak{g}$.

First, (C_{p-1}) gives

$$x_\mu\cdot z_T = x_\nu\cdot(x_\mu\cdot z_\Psi) + [x_\mu,x_\nu]\cdot z_\Psi.$$

Apply x_λ to both terms. For the second term, (C_{p-1}) gives

$$x_\lambda\cdot([x_\mu,x_\nu]\cdot z_\Psi) = [x_\mu,x_\nu]\cdot(x_\lambda\cdot z_\Psi) + [x_\lambda,[x_\mu,x_\nu]]\cdot z_\Psi.$$

For the first term, use (B_{p-2}) to write

$$x_\mu \cdot z_\Psi = z_\mu z_\Psi + w, \qquad w \in S_{p-2}.$$

The monomial $z_\mu z_\Psi$ has length $p-1$, its indices are at least ν, and $\nu < \lambda$. So (11.1) applies to the pair λ, ν and this monomial. Also (C_{p-1}) applies to w:

$$\begin{aligned} x_\lambda \cdot (x_\nu \cdot (z_\mu z_\Psi)) &= x_\nu \cdot (x_\lambda \cdot (z_\mu z_\Psi)) + [x_\lambda, x_\nu] \cdot (z_\mu z_\Psi), \\ x_\lambda \cdot (x_\nu \cdot w) &= x_\nu \cdot (x_\lambda \cdot w) + [x_\lambda, x_\nu] \cdot w. \end{aligned}$$

Adding the two lines,

$$x_\lambda \cdot (x_\nu \cdot (x_\mu \cdot z_\Psi)) = x_\nu \cdot (x_\lambda \cdot (x_\mu \cdot z_\Psi)) + [x_\lambda, x_\nu] \cdot (x_\mu \cdot z_\Psi).$$

Substituting both results,

$$\begin{aligned} x_\lambda \cdot (x_\mu \cdot z_T) = x_\nu \cdot (x_\lambda \cdot (x_\mu \cdot z_\Psi)) &+ [x_\lambda, x_\nu] \cdot (x_\mu \cdot z_\Psi) \\ &+ [x_\mu, x_\nu] \cdot (x_\lambda \cdot z_\Psi) + [x_\lambda, [x_\mu, x_\nu]] \cdot z_\Psi. \end{aligned}$$

The hypotheses are symmetric in λ and μ. Exchange them and subtract. The two middle terms are symmetric and cancel:

$$\begin{aligned} & x_\lambda \cdot (x_\mu \cdot z_T) - x_\mu \cdot (x_\lambda \cdot z_T) \\ &\quad = x_\nu \cdot \big(x_\lambda \cdot (x_\mu \cdot z_\Psi) - x_\mu \cdot (x_\lambda \cdot z_\Psi)\big) + \big([x_\lambda, [x_\mu, x_\nu]] - [x_\mu, [x_\lambda, x_\nu]]\big) \cdot z_\Psi \\ &\quad = x_\nu \cdot ([x_\lambda, x_\mu] \cdot z_\Psi) + [[x_\lambda, x_\mu], x_\nu] \cdot z_\Psi \\ &\quad = [x_\lambda, x_\mu] \cdot (x_\nu \cdot z_\Psi) \\ &\quad = [x_\lambda, x_\mu] \cdot z_T. \end{aligned}$$

The second equality uses (C_{p-1}) and the Jacobi identity in the form

$$[x_\lambda, [x_\mu, x_\nu]] = [[x_\lambda, x_\mu], x_\nu] + [x_\mu, [x_\lambda, x_\nu]].$$

The third uses (C_{p-1}) for the pair $[x_\lambda, x_\mu]$ and x_ν. Linearity in z gives (C_p). ∎

Proof of Theorem 11.6. Spanning is Proposition 11.5. Define the map $\rho\colon \mathfrak{g} \to \mathrm{End}(S)$ by

$$\rho(x) z = x \cdot z.$$

By (C) and linearity, ρ is a representation. Let $\tilde{\rho}$ be its extension to $U(\mathfrak{g})$ from Proposition 11.3. We claim

$$\tilde{\rho}(x_\Sigma) 1 = z_\Sigma.$$

Use induction on the length. For $\Sigma = \varnothing$ both sides are 1. For $\Sigma = (\lambda, \Sigma')$ we have $\lambda \le \Sigma'$. Since $\tilde{\rho}$ is a homomorphism, the induction hypothesis and (A) give

$$\tilde{\rho}(x_\Sigma) 1 = \rho(x_\lambda) \tilde{\rho}(x_{\Sigma'}) 1 = x_\lambda \cdot z_{\Sigma'} = z_\lambda z_{\Sigma'} = z_\Sigma.$$

If $\sum c_\Sigma x_\Sigma = 0$, apply $\tilde{\rho}$ of this element to 1. Then $\sum c_\Sigma z_\Sigma = 0$. The monomials z_Σ are independent, so every c_Σ is zero. ∎

11.4 Consequences

Corollary 11.9. *The map ι is injective. The ordered monomials of length at most p form a basis of U_p. Hence $\dim U_p = \binom{n+p}{p}$, the number of monomials of degree at most p in n variables.*

Proof. The elements $\iota(x_i)$ are distinct ordered monomials of length one, so they are independent. The second statement combines Proposition 11.5 with Theorem 11.6. ∎

From now on $\mathfrak{g}$ is identified with $\iota(\mathfrak{g})$.

Next we separate the leading part of a product from its lower-order corrections. By the reordering lemma, those corrections lie in lower pieces of the filtration. The associated graded algebra discards them, and PBW identifies what remains with a polynomial algebra. We use this comparison in §11.5 and in Sheet 11. The highest-weight construction of later lectures does not need it.

The *associated graded algebra* is

$$\operatorname{gr} U(\mathfrak{g}) = \bigoplus_{p\geq 0} U_p/U_{p-1}, \qquad (a+U_{p-1})(b+U_{q-1}) = ab + U_{p+q-1},$$

for $a \in U_p$ and $b \in U_q$. The product is well defined because $U_{p-1}U_q$ and U_pU_{q-1} lie in U_{p+q-1}. The *symmetric algebra* $\operatorname{Sym}(\mathfrak{g})$ is $T(\mathfrak{g})$ modulo the ideal generated by all $x \otimes y - y \otimes x$. The chosen basis of $\mathfrak{g}$ identifies $\operatorname{Sym}(\mathfrak{g})$ with S, by $x_i \mapsto z_i$.

Theorem 11.10. *The map $z_i \mapsto x_i + U_0$ extends to an isomorphism of graded algebras*

$$\psi\colon \operatorname{Sym}(\mathfrak{g}) \longrightarrow \operatorname{gr} U(\mathfrak{g}), \qquad z_\Sigma \longmapsto x_\Sigma + U_{p-1} \quad (\Sigma \text{ of length } p).$$

Proof. For $x, y \in \mathfrak{g}$, the commutator of $x + U_0$ and $y + U_0$ in $\operatorname{gr} U(\mathfrak{g})$ is

$$[x, y] + U_1 = 0 \quad \text{in } U_2/U_1,$$

since $[x, y] \in U_1$. By Proposition 11.5, the classes $x_i + U_0$ generate $\operatorname{gr} U(\mathfrak{g})$. These generators commute, so $\operatorname{gr} U(\mathfrak{g})$ is commutative. The polynomial algebra S is free commutative on the z_i. Hence there is an algebra homomorphism ψ with the displayed values.

By Corollary 11.9, the classes of ordered monomials of length exactly p form a basis of U_p/U_{p-1}. Thus ψ maps the monomial basis of degree p bijectively onto a basis of U_p/U_{p-1}. ∎

Corollary 11.11 (Factorisation)**.** *Let $\mathfrak{a}, \mathfrak{k} \subseteq \mathfrak{g}$ be subalgebras with $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{k}$ as vector spaces. The multiplication map*

$$U(\mathfrak{a}) \otimes U(\mathfrak{k}) \longrightarrow U(\mathfrak{g}), \qquad u \otimes v \longmapsto uv,$$

is a linear isomorphism. For any subalgebra $\mathfrak{a} \subseteq \mathfrak{g}$, the induced map $U(\mathfrak{a}) \to U(\mathfrak{g})$ is injective.

Proof. The universal property gives the maps $U(\mathfrak{a}) \to U(\mathfrak{g})$ and $U(\mathfrak{k}) \to U(\mathfrak{g})$ induced by inclusion. Order a basis of $\mathfrak{g}$ by listing a basis of $\mathfrak{a}$ first and then a basis of $\mathfrak{k}$. Each ordered monomial of $\mathfrak{g}$ is uniquely a product

$$x_\Sigma = x_{\Sigma'}x_{\Sigma''},$$

where $x_{\Sigma'}$ is an ordered monomial of $\mathfrak{a}$ and $x_{\Sigma''}$ one of $\mathfrak{k}$. By PBW for $\mathfrak{a}$ and $\mathfrak{k}$, the tensors $x_{\Sigma'} \otimes x_{\Sigma''}$ form a basis of $U(\mathfrak{a}) \otimes U(\mathfrak{k})$. The multiplication map sends this basis bijectively onto the PBW basis of $U(\mathfrak{g})$.

For any subalgebra, extend its ordered basis to an ordered basis of $\mathfrak{g}$. PBW shows that the induced map sends distinct basis monomials to distinct basis monomials. ∎

Now let $\mathfrak{g}$ be semisimple. Choose a Cartan subalgebra $\mathfrak{h}$ and positive roots as in Lecture 9, and put

$$\mathfrak{n}_{\pm} = \bigoplus_{\alpha \in R_+} \mathfrak{g}_{\pm\alpha}.$$

The root decomposition gives $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$, the *triangular decomposition*. The rules $[\mathfrak{h}, \mathfrak{g}_\alpha] \subseteq \mathfrak{g}_\alpha$ and $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$ of Lecture 7 show that $\mathfrak{n}_\pm$ and $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+$ are subalgebras. Corollary 11.11, applied to $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{b}$ and to $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+$, gives

$$U(\mathfrak{n}_-) \otimes U(\mathfrak{h}) \otimes U(\mathfrak{n}_+) \xrightarrow{\cong} U(\mathfrak{g}), \qquad U(\mathfrak{n}_-) \otimes U(\mathfrak{b}) \xrightarrow{\cong} U(\mathfrak{g}).$$

The second isomorphism makes $U(\mathfrak{g})$ a free right $U(\mathfrak{b})$-module with basis the ordered monomials of $\mathfrak{n}_-$. Lecture 12 uses this to construct induced modules. Since $\mathfrak{h}$ is abelian, the defining relations of $U(\mathfrak{h})$ and $\mathrm{Sym}(\mathfrak{h})$ coincide, so $U(\mathfrak{h}) = \mathrm{Sym}(\mathfrak{h})$.

11.5 Reordering in $U(\mathfrak{sl}_2)$

Order the basis of $\mathfrak{sl}_2$ as (f, h, e), matching $\mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$. The ordered monomials are $f^a h^b e^c$. The brackets $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$ of Lecture 6 give the rewriting rules

$$eh = he - 2e, \qquad hf = fh - 2f, \qquad ef = fe + h.$$

Figure 11.1 showed two routes for the word ehf. The next example writes out its left route line by line.

Example 11.12 (The word ehf)**.** Move e to the right first:

$$\begin{aligned} ehf &= (he - 2e)f \\ &= h(ef) - 2ef \\ &= h(fe + h) - 2(fe + h) \\ &= (hf)e + h^2 - 2fe - 2h \\ &= (fh - 2f)e + h^2 - 2fe - 2h \\ &= fhe - 4fe + h^2 - 2h. \end{aligned}$$

PBW gives uniqueness of this ordered expansion.

Now compare with $\mathrm{gr}\, U(\mathfrak{sl}_2)$. Write z_f, z_h, z_e for the variables of $\mathrm{Sym}(\mathfrak{sl}_2)$. The word ehf lies in U_3, and the correction terms lie in U_2. In U_3/U_2,

$$ehf + U_2 = fhe + U_2 = \psi(z_f z_h z_e).$$

In the associated graded algebra, the reordering is plain commutativity. In $U(\mathfrak{sl}_2)$ itself,

$$ehf - fhe = -4fe + h^2 - 2h \neq 0$$

by PBW independence. The term $-2h$ lies in U_1, so

$$ehf - fhe + U_1 = \psi(-4 z_f z_e + z_h^2) \neq 0 \quad \text{in } U_2/U_1.$$

11.6 Coproduct and antipode

This section is a preview of Lecture 14. There the tensor products and duals of modules are first defined by formulas and then recovered from the two maps below. Lecture 12 and Lecture 13 do not use these maps.

The algebra $U(\mathfrak{g}) \otimes U(\mathfrak{g})$ has product $(a \otimes b)(c \otimes d) = ac \otimes bd$. The *opposite algebra* $U(\mathfrak{g})^{\mathrm{op}}$ has product $a * b = ba$. An *antihomomorphism* is a linear map $S\colon U(\mathfrak{g}) \to U(\mathfrak{g})$ that satisfies $S(1) = 1$ and $S(uv) = S(v)S(u)$.

Proposition 11.13. *There is a unique algebra homomorphism $\Delta\colon U(\mathfrak{g}) \to U(\mathfrak{g}) \otimes U(\mathfrak{g})$ and a unique antihomomorphism $S\colon U(\mathfrak{g}) \to U(\mathfrak{g})$ such that for $x \in \mathfrak{g}$*

$$\Delta(x) = x \otimes 1 + 1 \otimes x, \qquad S(x) = -x.$$

Proof. Both statements follow from Proposition 11.2 once the bracket condition is checked. For Δ,

$$(x \otimes 1 + 1 \otimes x)(y \otimes 1 + 1 \otimes y) = xy \otimes 1 + x \otimes y + y \otimes x + 1 \otimes xy.$$

Exchange x and y and subtract. The two middle terms cancel:

$$[\Delta(x), \Delta(y)] = [x, y] \otimes 1 + 1 \otimes [x, y] = \Delta([x, y]).$$

For S, an antihomomorphism is a homomorphism into $U(\mathfrak{g})^{\mathrm{op}}$. Apply Proposition 11.2 to the map $\mathfrak{g} \to U(\mathfrak{g})^{\mathrm{op}}$, $x \mapsto -x$. Its bracket condition uses the opposite product:

$$(-x) * (-y) - (-y) * (-x) = (-y)(-x) - (-x)(-y) = yx - xy = -[x, y]. \qquad \blacksquare$$

Lecture 14 defines tensor products and duals of modules by direct formulas, and recovers those actions with Δ and S.

Self-checks

(1) Write eh^2 in the basis $f^a h^b e^c$ of $U(\mathfrak{sl}_2)$.

(2) List a basis of $U_2 \subseteq U(\mathfrak{sl}_2)$ and state its dimension.

(3) Express $S(ef)$ in the basis $f^a h^b e^c$.

Answers. (1) Use $eh = (h - 2)e$ twice:

$$eh^2 = (h - 2)eh = (h - 2)^2 e = h^2 e - 4he + 4e.$$

(2) The ordered monomials $1, f, h, e, f^2, fh, fe, h^2, he, e^2$ form a basis. The dimension is $\binom{5}{2} = 10$. (3) $S(ef) = S(f)S(e) = (-f)(-e) = fe$, already ordered.

Summary

The enveloping algebra $U(\mathfrak{g})$ turns brackets into commutators. Its unital modules are exactly the $\mathfrak{g}$-modules.

Reordering proves that the ordered monomials span $U(\mathfrak{g})$. Their independence comes from a polynomial module whose construction uses the Jacobi identity. Two consequences are $\operatorname{gr} U(\mathfrak{g}) \cong \operatorname{Sym}(\mathfrak{g})$ and, for semisimple $\mathfrak{g}$, $U(\mathfrak{g}) \cong U(\mathfrak{n}_-) \otimes U(\mathfrak{b})$ as vector spaces. The second is the input for Verma modules in Lecture 12.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 17.1–17.4; the independence proof follows Section 17.4. Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd edition, Chapter 9.